

**Limited labels, unseen continents: A low-cost embedding
workflow for cross-national alluvial gold-mine detection and
downstream river monitoring**

Entao Wang

This dissertation is submitted in partial fulfillment
of the requirements for the degree of MSc
in the Centre for Advanced Spatial Analysis,
Bartlett Faculty of the Built Environment, UCL

CASA0010
10,500 words
Sept 4, 2026
Supervised by Dr Ollie Ballinger

Abstract

Alluvial gold mining is dispersed, frequently unrecorded and costly to monitor, and supervised detection normally demands extensive manual labelling. This dissertation tests whether AlphaEarth Satellite Embeddings can support cross-national alluvial-mine detection and downstream river monitoring under a constrained computational budget. A gradient-boosted classifier was fitted on 35,275 labelled pixels from 191 polygons in five South American countries, then applied without retraining, at a threshold fixed on spatially buffered training folds, to seven held-out countries on three continents. F1 was 0.886 within the training region, 0.833 across four held-out South American countries and 0.873 across Ghana, Indonesia and Myanmar, although country values ranged from 0.772 to 0.954 and did not follow geographical distance. In a bounded Tapajós system, adding signed 2019–2024 embedding change to river and geographical context raised out-of-catchment R^2 from 0.412 to 0.626, but contributed less once historical mining was known. AlphaEarth therefore supports low-cost screening rather than autonomous mapping.

Key Words: Alluvial mining; Geospatial foundation models; Spatial transferability; Downstream river monitoring

Declaration

I hereby declare that this dissertation is all my own original work and that all sources have been acknowledged. It is 10,500 words in length.

Content

Abstract	2
List of Figures.	5
List of Tables.	5
List of Abbreviations	6
1 Introduction	8
2 Literature Review	11
3 Data, Study Area	20
4 Methodology.	23
5 Results	32
6 Discussion	38
7 Conclusion	44
Code availability	44
References	45

List of Figures

Figure 2.1 Alluvial gold mining in Ghana

Figure 2.2 Components of an alluvial gold-mining site

Figure 3.1 Study areas: 15 areas of interest across three continents

Figure 4.1 Workflow for alluvial-mine classification and geographically independent evaluation

Figure 4.2 Workflow for the downstream river-change analysis

Figure 5.1 Structure of the AlphaEarth embedding space: PCA and UMAP projections of the labelled training pixels

Figure 5.2 Transfer to seven held-out countries

Figure 5.3 Agreement with Amazon Mining Watch, four held-out countries

List of Tables

Table 2.1 Functional components of an alluvial gold-mining site

Table 5.1 Classifier performance under training-region spatial validation

Table 5.2 Performance in the seven held-out countries

Table 5.3 Agreement with Amazon Mining Watch at sampled locations

Table 5.4 Out-of-catchment prediction of upstream mining growth

List of Abbreviations

AMW	Amazon Mining Watch
AOI	Area of Interest
ASM	Artisanal and Small-Scale Mining
ASGM	Artisanal and Small-Scale Gold Mining
BC	Before Christ
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
EECU	Earth Engine Compute Unit
EITI	Extractive Industries Transparency Initiative
F1	Harmonic mean of precision and recall
GDP	Gross Domestic Product
GeoJSON	Geographic JavaScript Object Notation
GeoTIFF	Georeferenced Tagged Image File Format
GRI	Global Reporting Initiative
HydroSHEDS	Hydrological Data and Maps Based on Shuttle Elevation Derivatives at Multiple Scales
IFC	International Finance Corporation
IGF	Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development
ILO	International Labour Organization
IRP	International Resource Panel
JRC	Joint Research Centre
MAE	Mean Absolute Error
MODIS	Moderate Resolution Imaging Spectroradiometer
NDTI	Normalised Difference Turbidity Index
OECD	Organisation for Economic Co-operation and Development
PCA	Principal Component Analysis
PFAF_ID	Pfaffstetter Basin Identifier
PR-AUC	Area Under the Precision–Recall Curve
R²	Coefficient of Determination
RIV_ORD	River-Order Attribute
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SMOTE	Synthetic Minority Over-sampling Technique

SWIR	Short-Wave Infrared
UMAP	Uniform Manifold Approximation and Projection
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
WHO	World Health Organization
XGBoost	Extreme Gradient Boosting

Limited labels, unseen continents: A low-cost embedding workflow for cross-national alluvial gold-mine detection and downstream river monitoring

1 Introduction

Archaeological evidence demonstrates that humans were deliberately extracting mineral resources by at least c. 10,000 BC (Salazar et al., 2011). Since then, human societies have non-stopped extracted resources from the Earth, including resources such as coal, sand, clay and metals. The Industrial Revolution and subsequent industrialisation greatly accelerated mineral production, expanding the extractive frontier across the world. According to Maus et al. 2020, there are 14.2 million acres, of visible surface-mining land currently existing on our planet. From the petrochemical feedstocks used in plastic bags to the uranium and metals required for nuclear reactors, modern societies depend on extracted materials in almost every sphere of production and consumption (GRI, 2024). Global natural-resource extraction has tripled over the past five decades and, if without concerted intervention, could increase by a further 60% between 2020 and 2060 (UNEP and IRP, 2024). Speaking from economical perspective, the value of all mining products was equivalent to approximately 1.2% of global GDP in 2016, after peaking at 1.9% in 2011 (Ericsson and Löf, 2019). The sector can be considerably more important in mineral-dependent economies in the Global South. In Guinea, for example, mining generates approximately 15% of GDP and 80% of export earnings (World Bank Group, 2021).

Although mining makes a substantial contribution to human society and economic development, it also poses major threats to the environment. A recent global assessment estimated that mining directly caused approximately 12,865 km² of forest loss between 2001 and 2020, equivalent to an annual average of approximately 643 km²(Ren et al., 2026). This estimate is likely to be conservative because global mining inventories remain incomplete, while informal and unregistered operations are often poorly represented in official records.

As awareness of the need to regulate mining grew, methods for monitoring mining activities developed alongside the expansion of the industry. Historically, the quantification of mining activity depended largely on field surveys, administrative records and the manual interpretation of aerial or satellite imagery. Manual digitisation can produce highly accurate local mine maps,

but it is labour-intensive and difficult to apply consistently over extensive geographical areas (Nursamsi et al., 2024). In recent decades, the increasing availability of satellite imagery and unmanned aerial equipment has made it possible to observe and monitor mining activities remotely, repeatedly and over much larger areas (Loots, Grobbelaar and Van der Lingen, 2022). Since the 2000s, advances in machine learning have further facilitated the automated detection, characterisation and monitoring of mines using remote-sensing data (Thiruchittampalam et al., 2025).

Despite advances in mining regulation and monitoring, several contemporary challenges necessitate the continued refinement of monitoring techniques. First, producing labelled imagery for supervised machine learning requires extensive manual effort. For example, Maus et al. (2020) visually interpreted satellite imagery around more than 6,000 active mining sites worldwide and manually delineated 21,060 mining-area polygons. Similarly, Tang and Werner (2023) manually delineated 74,548 mine-area polygons across 135 countries and regions. Second, conventional inputs derived from a single satellite acquisition or an isolated spectral index provide only a partial representation of surface conditions. Although individual satellite images contain spatial information, they do not inherently summarise longer-term temporal trajectories or integrate information from multiple sensors and the broader spatial context surrounding each pixel (Brown et al., 2025; Google Earth Engine, 2026). Moreover, conventional machine-learning workflows for mining monitoring can require substantial data preprocessing, resulting in considerable computational and labour costs. To capture complementary optical, radar, topographic and temporal characteristics, researchers may need to retrieve, cloud-mask, composite, co-register, resample and combine observations from multiple sources. This complexity is evident in mining studies that have constructed optical and radar Sentinel time series, combined Sentinel-1 and Sentinel-2 imagery with elevation data, or processed long Landsat time series and derived spectral indices to monitor mining disturbance and reclamation (Jabłońska et al., 2024; Ofori-Ampofo et al., 2025; Xiao et al., 2023).

AlphaEarth Foundations (published by Google in July 2025), offers several advantages for environmental mapping. It is particularly relevant to applications in which reliable labelled data are scarce. According to research done by Brown et al. (2025), across a diverse suite of

classification, regression and change-detection evaluations, AlphaEarth reduced the overall error magnitude by approximately 23.9% relative to the next-best method and it retained this performance advantages under highly data-scarce conditions.

Moreover, rather than relying solely on individual satellite observations or manually designed spectral indices, AlphaEarth integrates spatial, temporal and measurement information from multiple Earth-observation sources into a unified representation (Google DeepMind, 2025). The associated Satellite Embedding dataset represents each 10 m pixel using a 64-dimensional vector that summarises annual surface conditions at and around that location. This analysis-ready feature reduce the need for extensive image compositing, cloud handling and manual feature engineering, while mitigating some problems associated with missing observations and sensor artefacts (Google Earth Engine, 2026). Its globally consistent feature space may also provide a suitable basis for investigating whether classifiers trained using geographically limited mining labels can be transferred to previously unseen countries.

Against this background, this research focuses on globally important alluvial-mining regions in the Amazon Basin, West Africa and Southeast Asia. It investigates whether AlphaEarth can support the cross-national identification of alluvial mining and the monitoring of downstream river changes associated with upstream mining activity, while keeping computational requirements at a relatively low level. The central research question is therefore: Can AlphaEarth support a computationally efficient workflow for detecting alluvial mining across countries and identifying downstream river signals associated with upstream mining expansion?

To answer this research question, the study has two objectives:

1. To evaluate whether a model trained using geographically limited AlphaEarth samples can identify alluvial mining in previously unseen countries, including countries located on different continents.
2. To assess whether temporal changes in downstream river AlphaEarth embeddings contain information about hydrologically accumulated upstream mining expansion beyond that provided by river morphology and geographical context.

Addressing these objectives may provide government agencies, research institutions and other organisations with a computationally economical approach to updating mine detection and identifying downstream river changes associated with upstream mining activity. The relatively low data-processing requirements of the proposed AlphaEarth workflow may be particularly valuable for resource-constrained organisations and countries in the Global South, where access to extensive computing infrastructure and specialised monitoring personnel may be limited. Ultimately, this research seeks to contribute to improving the monitoring and management of mining activities, particularly in regions where existing mining inventories and environmental-monitoring capacity remain incomplete.

2. Literature Review

2.1 What is Alluvial mining

Alluvial mining refers to the extraction of valuable minerals, particularly gold, from unconsolidated sediments deposited by rivers and other fluvial processes, typically within riverbeds, floodplains and terraces (Mathioudakis et al., 2023). Originally, the deposits from water are formed through natural hydraulic sorting. As flowing water loses energy, denser particles settle and accumulate in river channels, bars, floodplains and abandoned terraces. The U.S. Geological Survey (2025) describes these mineral-rich deposits as alluvial placers.



Fig 2.1 Alluvial Mining in Ghana (Boamah, 2016)

Mining process reproduces this sorting process over a much shorter timescale. Sediment is excavated and mixed with water, allowing gravity to separate the denser mineral fraction from lighter material. This process has two important implications for the spatial pattern of alluvial mining. First, mining usually occurs close to present or former river channels because these are the areas where valuable sediments have naturally accumulated. Second, the extraction process relies on water to transport and separate sediment, which keeps mining closely connected to the river system. Dethier et al. (2023) incorporate this relationship into their remote-sensing definition of alluvial mining. They define it as the excavation of river or floodplain sediment with a direct surface-water connection to a channel. These spatial pattern make detection of alluvial mining from remote sensing image possible.

Alluvial mining is not the same as artisanal and small-scale mining (ASM). The term alluvial describes the type of deposit and the physical setting in which extraction occurs. In contrast, artisanal and small-scale mining refers to modes of organisation, capital and technology, although national definitions of ASM remain inconsistent (Hentschel, Hruschka and Priester, 2002; Fritz et al., 2018). This dissertation focuses on alluvial and river-connected gold mining. Much of this activity may be artisanal, small-scale or informal, but these characteristics cannot be reliably identified from spectral signatures alone. The analysis therefore does not use remote-sensing evidence to determine the legal status or operational scale of individual mining sites (Nursamsi et al., 2024).

2.2 Components of an alluvial mine

A typical alluvial gold mine includes facilities for excavation, size separation, gravity concentration and waste disposal. Mining usually begins with the removal of vegetation and overburden layers. Excavators, hydraulic jets or suction pumps are then used to mix sediment with water and create a slurry. Screens remove the coarser material before the remaining sediment passes through a sluice, jig or centrifugal concentrator. These devices retain the denser gold particles, while lighter material becomes tailings (UNEP, 2012; Veiga and Gunson, 2020). Gold may then undergo further processing using mercury amalgamation, and finally, settling ponds can be used to recover and recycle water (Esdaile and Chalker, 2018). During

this stage, some poorly contained tailings and discharge flows can create a direct connection between the mining site and nearby rivers.

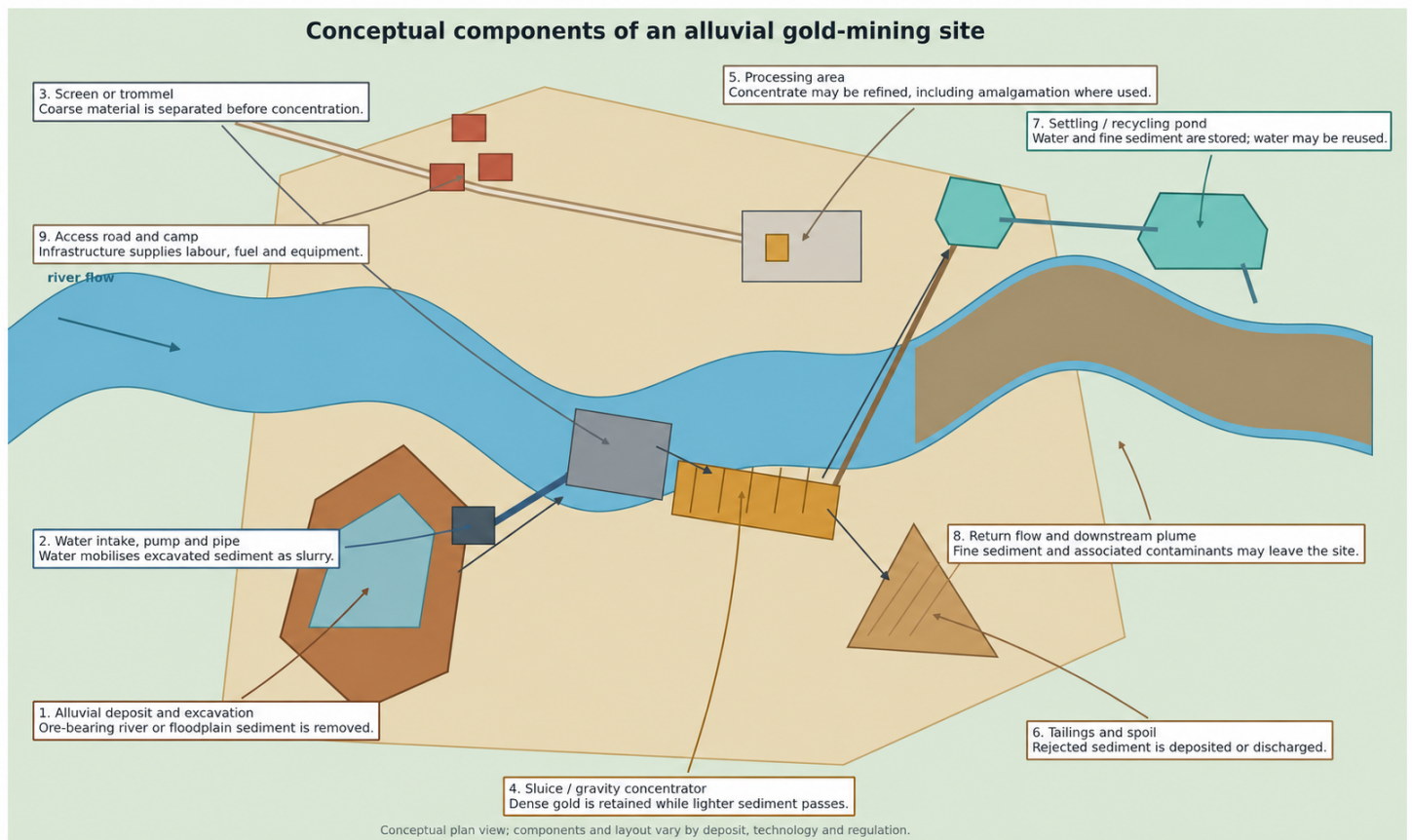


Fig. 2.2 Components of an alluvial gold-mining site. Schematic based on U.S. Geological Survey (2025), UNEP (2012), Veiga and Gunson (2020) and Dethier et al. (2023). The arrangement is illustrative: individual sites need not contain every component. See below table for detailed explanation for each component.

Table 2.1 Functional components and remote-sensing expressions of an alluvial gold-mining site

Component	Operational role
1. Deposit and excavation zone	Supplies ore-bearing river or floodplain sediment
2. Pump, pipe and water diversion	Mobilises sediment and supplies processing water
3. Screen or trommel	Removes coarse gravel and standardises feed size
4. Sluice or gravity concentrator	Retains dense gold and rejects lighter sediment

5. Processing area	Refines concentrate; mercury amalgamation may occur
6. Tailings and spoil	Stores rejected coarse and fine material
7. Settling or recycling pond	Settles fines and allows water reuse
8. Return flow and downstream plume	Carries residual sediment and associated contaminants away
9. Access and camp infrastructure	Moves labour, fuel, equipment and gold

2.3 Environmental impacts of alluvial mining

Alluvial mining can remove vegetation, soil and channel sediment from the mining area. In Madre de Dios, two major mining sites converted approximately 6,600 ha of primary forest and wetlands between 2003 and 2009. Annual conversion also increased from 292 ha to 1,915 ha between the first and second halves of this period (Swenson et al., 2011). Higher-resolution monitoring later identified many sub-hectare clearings that had previously been missed (Asner et al., 2013). A subsequent assessment estimated that mining caused 95,751 ha of forest loss in south-eastern Peru between 1985 and 2017 (Caballero Espejo et al., 2018). Across tropical moist forests in South America, approximately 1,680 km² of forest was lost around gold-mining sites between 2001 and 2013. Almost one-third of this loss occurred within 10 km of protected areas (Alvarez-Berrios and Aide, 2015). Although these studies use different spatial extents and definitions, they show how many relatively small mining sites can produce substantial cumulative forest loss. Additional indirect impacts may also result from the construction of roads, settlements and other supporting infrastructure (Sonter et al., 2017).

Alluvial mining also changes riverbanks and releases large amounts of fine sediment or harmful chemical substances into surrounding water systems. In Madre de Dios, mining increased downstream suspended-sediment concentrations and reversed their natural seasonal pattern (Dethier, Sartain and Lutz, 2019). At the global scale, 140 of 173 large tropical rivers affected by mining showed post-mining sediment concentrations that were more than twice their pre-mining levels. These impacts extended across more than 35,000 km of river channel (Dethier et al., 2023). Fine sediment can reduce light availability, bury aquatic habitats and negatively affect fish populations (Kemp et al., 2011). It can also transport contaminants attached to

sediment particles. In Ecuador's Puyango basin, mining tailings were associated with reduced aquatic biodiversity (Tarras-Wahlberg et al., 2001). In Madre de Dios, mining-related mercury has been detected in sediment, suspended solids and fish hundreds of kilometres downstream from mining areas (Diringer et al., 2015).

Mercury use creates an additional pathway of environmental and human exposure. Mercury released during amalgam burning can enter the atmosphere and later move through terrestrial and aquatic food webs. Exposure to mercury can cause neurological, renal and immunological effects (Gibb and O'Leary, 2014; WHO, 2016). Deforestation can also release mercury stored in soils (Diringer et al., 2020), while nearby forests can capture mercury emitted into the atmosphere (Gerson et al., 2022). However, artisanal and small-scale mining also provides an important source of income in regions where alternative livelihoods may be limited (Fritz et al., 2018; Verbrugge and Geenen, 2020). Environmental monitoring should therefore remain separate from assumptions about the legal status of mining activities or the motivations of individual miners.

2.4 Global extent and environmental inequality

The scale of the sector remains uncertain because a large proportion of mining activity is informal and therefore poorly recorded. The ILO estimated that 13 million people were engaged in small-scale mining in 1999, while a later synthesis estimated that 40.5 million people were directly involved in ASM in 2017 (ILO, 1999; Fritz et al., 2018). However, these estimates include a wide range of commodities and mining methods and are not limited to alluvial gold mining.

The importance of this monitoring problem is particularly clear in countries across the Global South, where mining industry is often a major contributor for local economy. In sub-Saharan Africa, small-scale mining is strongly linked to subsistence agriculture and provides an important source of non-farm income for many rural households (Hilson, 2016). However, studies across Africa have also documented widespread environmental and health impacts associated with artisanal gold mining, particularly from mercury and other potentially toxic elements (Ondayo et al., 2024). Similar patterns are reported in Latin America, where gold

mining has expanded across parts of the Amazon and is associated with forest disturbance, water contamination and informal production systems (Mestanza-Ramón et al., 2022). In Southeast Asia, more than 2,000 ASGM locations have been reported across 30 Indonesian provinces, providing employment and income for more than two million people, while many operations remain unlicensed and use relatively simple processing technologies (Meutia, Lumowa and Sakakibara, 2022). More broadly, recent reviews show that ASGM research remains concentrated in sub-Saharan Africa, South America and parts of Southeast Asia, but much of the evidence is still based on local or national case studies (Agyapong et al., 2026). These conditions strengthen the need for monitoring approaches that can identify dispersed mining activity consistently across large and geographically diverse areas.

2.5 Mining policy and the Sustainable Development Goals

The management of mining activities is directly relevant to Sustainable Development Goals (SDGs) 3, 6, 12, 15 and 16. SDG 3 concerns health risks associated with occupational hazards and exposure to mining pollutants. SDG 6 addresses the protection of water quality from sediment, mercury and other contaminants. SDG 12 promotes responsible mineral production, resource use and waste management. SDG 15 concerns the protection of terrestrial ecosystems from deforestation, soil degradation and habitat loss. And finally SDG 16 supports effective institutions, transparent licensing and the enforcement of mining regulations (United Nations General Assembly, 2015; UNDP et al., 2016). Mining governance should therefore integrate environmental monitoring, pollution control, transparent production records and effective regulatory enforcement.

International instruments manage mining activities through pollution control, production transparency and corporate responsibility. The Minamata Convention focuses specifically on mercury pollution from ASGM. Article 7 and Annex C require affected countries to prepare national action plans, estimate mercury use, eliminate the most harmful practices, reduce human exposure and support the formalisation of mining activities (Minamata Convention on Mercury, 2013).

The OECD applies risk-based due diligence to mineral supply chains, while the EITI Standard requires countries to estimate ASM production and explain limitations in their monitoring

systems (OECD, 2016; EITI, 2023). The IFC and IGF provide additional principles for environmental protection, labour conditions, community engagement and mine closure (IFC, 2012; IGF, 2023). However, formalisation remains difficult because miners may face high costs, insecure land tenure and inaccessible licensing procedures. Regulatory interventions may therefore be ineffective without the participation of miners and sufficient financial support (Prescott et al., 2022).

2.6 Mining detection methods: from manual interpretation to transferable models

Mine mapping initially relied on field inventories, administrative records and visual interpretation of aerial or satellite imagery. Human interpreters can identify mines using features such as shape, texture, position relative to rivers and the surrounding forest. Manual interpretation still forms the basis of several large global mining datasets (Maus et al., 2022; Tang and Werner, 2023). However, manual digitisation is time-consuming, difficult to update and often biased towards mining sites that are already known or visually distinctive. Automated approaches reduce this burden, but they do not remove the need for labelled data. Human interpretation is still required to produce training and validation samples.

Semi-automatic approaches commonly combine Landsat or Sentinel-2 reflectance with vegetation, water and bare-soil indices, change detection and expert-defined rules. These methods have been used to map mining in Madre de Dios, reconstruct several decades of mining activity in the Tapajós region and detect small mining areas in wet tropical environments (Elmes et al., 2014; Lobo et al., 2016; Isidro et al., 2017). Sentinel-2 can improve the delineation of relatively small mining features because of its finer spatial resolution (Lobo et al., 2018). Object-based texture measures can provide information about the surrounding spatial context (Simionato et al., 2021), while Landsat time series can help distinguish persistent mining disturbance from temporary areas of bare ground (Fonseca, Marshall and Salama, 2024). These approaches are relatively interpretable, but they still require substantial preprocessing. This can include image compositing, cloud masking, feature engineering and the selection of region-specific thresholds.

Machine-learning approaches use labelled examples to learn relationships between remote-sensing features and mining classes. Random forests are widely used because they can handle large numbers of mixed predictor variables, while XGBoost can model more complex interactions between features (Belgiu and Drăguț, 2016; Chen and Guestrin, 2016). Convolutional neural networks can also learn spatial patterns directly from imagery. They have been used to detect shallow mining areas in Ghana, segment mines and tailings, and map alluvial mining change in the Amazon (Gallwey et al., 2020; Balaniuk, Isupova and Reece, 2020; Camalan et al., 2022; Couttenier et al., 2022). Cloud-computing platforms and candidate-area screening have made large-area analysis more practical, while radar data can provide additional information in regions affected by persistent cloud cover (Gorelick et al., 2017; Becerra et al., 2024; Ferreira Neto et al., 2024). However, research on ASM remains geographically concentrated. It therefore remains unclear how well classifiers developed in one landscape can perform in geographically different mining regions (Nursamsi et al., 2024).

AlphaEarth provides a different approach to this problem. Its annual Satellite Embedding dataset represents each 10 m location using a globally consistent 64-dimensional feature vector that combines spatial, temporal and multi-source Earth-observation information (Brown et al., 2025; Google Earth Engine and Google DeepMind, 2025). These analysis-ready features may reduce the need for extensive preprocessing and may also support classification when only limited labelled data are available. However, strong performance on general Earth-observation tasks does not necessarily demonstrate that mining classifiers can transfer between regions. A recent low-shot application used AlphaEarth to map mining within a forest reserve in Ghana, showing that mining applications are beginning to emerge but remain geographically localised (Nsiah, Laari and Guan, 2026). Evaluation design is therefore important. Nearby training and test pixels often share similar environmental and landscape characteristics, meaning that random data splits can produce overly optimistic estimates of model performance (Roberts et al., 2017; Ploton et al., 2020). Spatial cross-validation provides a stronger test because training and validation data are separated geographically. However, it measures the ability of a model to extrapolate to new locations rather than providing an unbiased estimate of total map accuracy, which requires probability-based reference sampling (Wadoux et al., 2021). No study identified in this review has evaluated alluvial-mine classification across completely held-out countries and continents. This gap motivates the use of site-balanced training and geographically independent evaluation in this dissertation.

2.7 Methods for assessing downstream impacts

Monitoring downstream mining impacts requires linking upstream mining to a downstream river response. Field studies commonly compare affected and unaffected catchments. In Suriname, Mol and Ouboter (2004) used a paired-watershed design; the mining-affected stream contained 318–2,468 mg/L of suspended sediment and had lower fish diversity. This provides direct evidence, but extensive sampling is difficult across remote tropical rivers.

Satellite observations provide repeatable coverage. Park and Latrubesse (2014) calibrated MODIS reflectance using eleven years of field measurements to model suspended-sediment patterns across the Amazon. Espinoza-Villar et al. (2018) similarly combined MODIS observations with 881 field samples along a 2,050 km section of the Solimões River. However, river reflectance also changes with discharge, season, channel morphology and sediment supplied by natural tributaries. Satellite-derived responses must therefore be interpreted within their hydrological context.

Hydrological position is important because geographical proximity does not demonstrate that a mine drains into a river reach. Mining exposure should instead be accumulated through connected upstream sub-basins. This dissertation applies this principle to a reverse monitoring question: whether changes in downstream AlphaEarth embeddings contain information about upstream mining expansion. Because these embeddings have no physical units, the relationship represents predictive association rather than direct evidence of turbidity, mercury contamination or causality.

2.8 Synthesis and Research Gap

The literature shows that alluvial mining creates visible but geographically variable patterns of excavation, tailings, ponds and altered rivers. Existing remote-sensing studies have expanded mining inventories, but most models are trained and evaluated within the same region. Their ability to transfer to unseen countries therefore remains uncertain. Downstream studies also generally begin with known mines and measure their subsequent river impacts, rather than asking whether downstream observations can reveal upstream expansion. Therefore, this dissertation evaluates whether AlphaEarth can support cross-country alluvial-mine detection

from limited labels and whether temporal changes in downstream river embeddings provide information about hydrologically connected upstream mining growth.

3. Data and Research Areas

3.1 Research Areas

This research examines alluvial gold-mining environments across South America, Southeast Asia and West Africa. The Amazon Basin and adjacent northern South American mining belts form the principal region, while Southeast Asia and West Africa test whether patterns learned in South America remain transferable.

The mine-detection experiment contains 15 areas of interest (AOIs). Each was a nominal 110 x 110 km square centred on a documented alluvial-mining district, giving a combined nominal extent of 181,500 km². Eight training AOIs cover three regions in Brazil, one in Peru, two in Colombia, one in Guyana and one in Ecuador. Bolivia, Venezuela, Suriname and French Guiana were withheld for same-continent testing. Ghana, Indonesia and Myanmar were reserved for cross-continent testing.

The downstream experiment was restricted to a manually bounded northern section of the Tapajos system. Processing its complete level-4 catchment, approximately 494,533 km², was computationally infeasible. The final window retained the main northern mining concentrations and was intersected with the catchment boundary, producing the largest continuous and hydrologically coherent area affordable within the available quota.

3.2 AlphaEarth Satellite Embeddings

The principal dataset was the AlphaEarth Satellite Embedding V1 annual product, accessed through Google Earth Engine as `GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL`. AlphaEarth integrates spatial, temporal and measurement context from multiple Earth-observation sources into an analysis-ready representation (Brown et al., 2025). Each 10 m pixel contains 64 values, named A00-A63, summarising annual surface conditions at and around that

location. These are learned latent axes rather than spectral bands: no dimension represents one physical property, and all 64 must be used collectively. The unit-normalised vectors are available annually from 2017 onwards (Google Earth Engine and Google DeepMind, 2025).

For mine detection, the complete 2024 vector was extracted at each labelled location and supplied directly to the classifiers. For the downstream experiment, the same river stations were sampled in 2019 and 2024, and the 64-dimensional difference vector represented temporal change. Thus, the study used AlphaEarth at its native 10 m spatial resolution and annual temporal resolution.

3.3 Supplementary Data

Sentinel-2 Surface Reflectance Harmonized imagery supplied the visual basis for manual interpretation. Its visible and near-infrared bands have 10 m native resolution, while the shortwave-infrared bands used here have 20 m resolution. Cached 2024 true-colour and SWIR composites supported labelling but did not enter the final classifier.

The Dethier et al. (2023) vector inventory of 396 river-mining districts, developed from a 37-year Landsat record at 30 m resolution, anchored AOI placement and site grouping. Amazon Mining Watch supplied mining-scar masks derived from 10 m Sentinel-2 imagery. Its 2019 and 2024 masks measured upstream mining growth, while the 2024 product supported an independent agreement assessment. As it is model-derived, agreement was treated as corroboration rather than ground-truth accuracy.

Hydrological structure came from HydroBASINS level 12 and the HydroSHEDS Free-Flowing Rivers network, both derived from approximately 500 m hydrographic data (Lehner and Grill, 2013; Grill et al., 2019). JRC Global Surface Water v1.4 provided 30 m water occurrence for 1984-2021 (Pekel et al., 2016). Landsat 8/9 Collection 2 Level 2 supplied 30 m red and green reflectance for NDTI in 2019 and 2024. Natural Earth Admin-0 map units at 1:10 million scale supplied country boundaries.

3.4 Computational Power Limit

The research used the Google Earth Engine, limited to 1,000 EECU-hours per month, and free Google Colab. Colab resources are temporary and not guaranteed (Google Colab, 2026). An early workflow calculated approximately 153 million pixels before discarding most of them and consumed 226,651 EECU-seconds. A river-sampling task timed out after 930,619 EECU-seconds, while a denser basin design consumed approximately 4.05 million EECU-seconds, exceeding the monthly allowance.

The final workflow therefore generated sample locations before reading AlphaEarth, cached Sentinel-2 basemaps, divided exports by country or into five chunks, averaged AlphaEarth within river-station corridors and aggregated NDTI directly by sub-basin. A complete 10 m classification of South America was not attempted. The final design represents the largest scientifically coherent regional analysis achievable with the available resources, rather than an exhaustive continental inventory.

3.5 Training and Test Region Selection Protocol

Training and test AOIs followed the same fixed-size rule and were centred on documented mining districts. Their coordinates were fixed before final labelling. Brazil, Peru, Colombia, Guyana and Ecuador formed the training pool. Five-fold spatial validation held out complete sites and removed training sites within 10 km of each test site. Bolivia, Venezuela, Suriname and French Guiana formed the same-continent holdout; Ghana, Indonesia and Myanmar formed the cross-continent holdout. The decision threshold was selected from spatially buffered training predictions and fixed before held-out evaluation.

3.6 Labelling Protocol

One analyst manually delineated labels in Google Earth Engine using 2024 Sentinel-2 composites. The principal positive definition covered excavation surfaces, tailings, ponds and disturbed ground. A secondary definition retained only unambiguous bare mining surfaces and tailings ponds. Negatives included agriculture, forest, grassland, urban or bare ground, clear rivers, and turbid or braided rivers. Difficult negatives were deliberately placed near mining districts.

Every MultiPolygon was separated into individual features and assigned a `polygon_id`, `site_id`, `country`, `class` and `data split`. Fragments belonging to the same mining district shared one `site_id`, preventing fragmented workings from receiving excessive weight. No polygon was rejected because of its area. Where possible, boundaries were inset by 10 m before sampling to reduce mixed edge pixels. Polygons above 5 ha contributed approximately 200 randomly located pixels, whereas smaller polygons were sampled exhaustively; analysis imposed a final maximum of 300 pixels per polygon. The same procedure was applied to training and test labels. The final principal dataset contained 35,275 pixels from 191 polygons and 148 sites across five training countries.

3.7 Platforms Used

Google Earth Engine and JavaScript were used for imagery access, labelling, spatial sampling and export. Google Drive stored the CSV, GeoJSON and GeoTIFF outputs.

Google Colab and Python were used for quality control, spatial validation, XGBoost modelling, raster processing, hydrological accumulation and figure production, principally through pandas, NumPy, GeoPandas, Rasterio, SciPy, scikit-learn, XGBoost and Matplotlib.

4. Methodology

4.1 Overall Analytical workflow

The methodology contained two related analyses. The first analysis examined whether alluvial mining could be identified from the 2024 AlphaEarth embeddings and whether a classifier trained in five countries remained effective in geographically separate countries. The second analysis examined whether changes in downstream river embeddings between 2019 and 2024 contained information about hydrologically connected upstream mining growth. These analyses used the same embedding product but answered different questions. Mine detection was a binary classification problem at the 10 m pixel level, whereas downstream monitoring was a continuous prediction problem at the HydroBASINS level-12 sub-basin level.

The workflow was designed around the available computational resources. Google Earth Engine generated spatial samples and read AlphaEarth only at those locations. Google Colab then completed data cleaning, model fitting, spatial validation and figure production. This avoided processing the 64-dimensional embedding across every pixel in each AOI. A random seed of 42 was used throughout the sampling and modelling stages to make the results reproducible.

4.2 Alluvial-mine classification

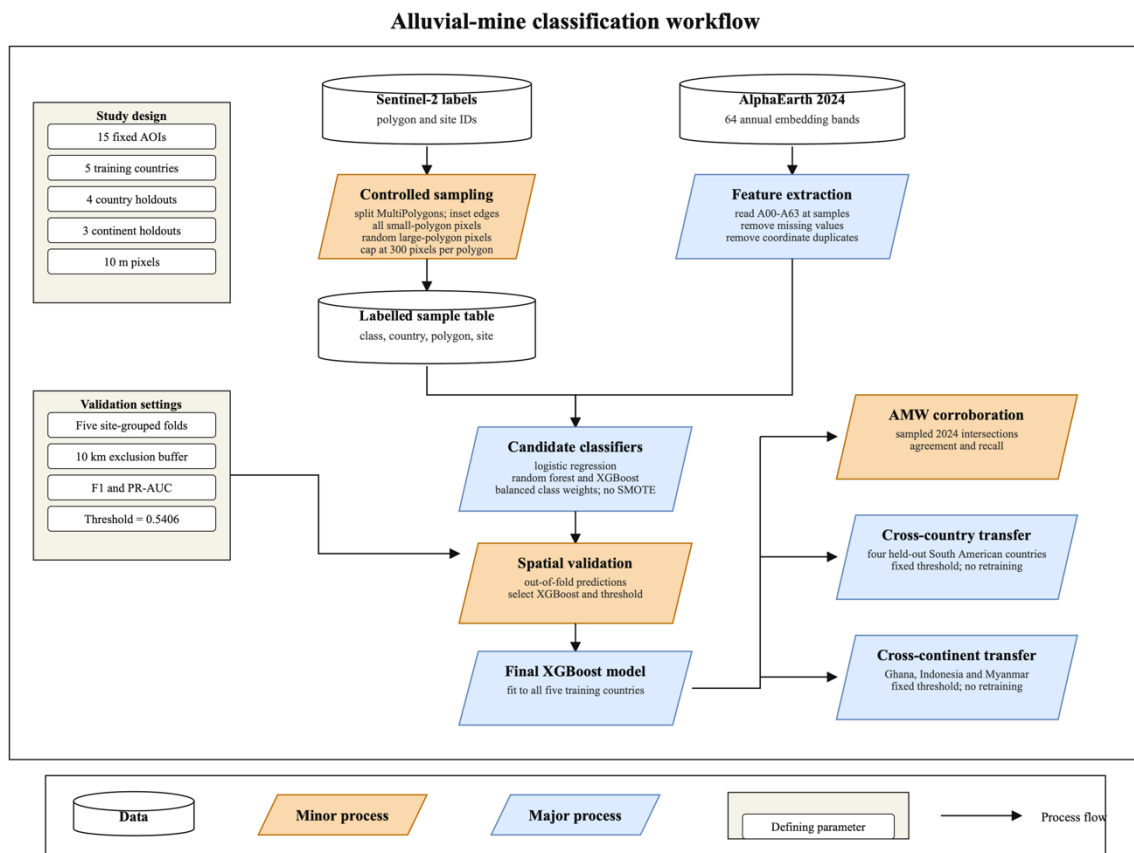


Figure 4.1. Workflow used for alluvial-mine classification and geographically independent evaluation.

4.2.1 Sample preparation

The classification analysis used the principal A_complex definition. Pixels inside the manually delineated mine complexes were assigned to the positive class. All eight non-mining classes formed the negative class, including turbid and braided rivers, forest, agriculture and urban bare ground. The alternative B_pure_surface definition was retained only for sensitivity

analysis because it represented individual mining surfaces rather than the complete mine complex.

For polygons larger than 5 ha, approximately 200 random sample locations were generated before AlphaEarth was read. Polygons at or below 5 ha were sampled exhaustively because they contained relatively few pixels and could otherwise be under-represented. A final cap of 300 observations per polygon was applied in Colab. This procedure reduced the influence of very large mine-complex polygons while retaining genuine georeferenced observations. It did not make the number of samples identical between countries or sites.

All 64 AlphaEarth variables, A00-A63, were required. Rows with missing values were removed, and duplicate observations were identified from rounded longitude and latitude coordinates and retained only once. Polygon and site identifiers were namespaced where necessary to prevent unrelated features in different exports from sharing the same identifier. After quality control, the principal training dataset contained 35,275 pixels from 191 polygons and 148 sites.

Synthetic oversampling was not used. Although SMOTE can increase the numerical representation of a minority class, a synthetic observation has no genuine geographical location and cannot be assigned reliably to a spatial validation fold. Class imbalance was instead addressed within the classifiers, while the polygon cap prevented a small number of large polygons from dominating the sample.

4.2.2 Classification models

Three classifiers were used to examine whether the result depended on model complexity. Logistic regression provided a linear baseline and tested whether one standardised decision boundary could separate mining from non-mining in the 64-dimensional feature space. It used balanced class weights and a maximum of 2,000 iterations. Random forest provided a conventional non-linear remote-sensing model (Belgiu and Drăguț, 2016). It contained 150

trees, used a minimum leaf size of five, trained each tree on 50% of the available observations and applied balanced class weights.

XGBoost was the principal classifier because it can represent non-linear relationships and interactions while remaining efficient on tabular data (Chen and Guestrin, 2016). It used 300 trees, a maximum depth of five, a learning rate of 0.08, row subsampling of 0.8, column subsampling of 0.6 and L2 regularisation of 1.0. Trees were fitted using the histogram method. Within each training fold, `scale_pos_weight` was set to the ratio between negative and positive observations. The same model settings were used in every geographical evaluation, and no model was re-tuned for an individual test country.

4.2.3 Training-region self spatial validation

The first evaluation measured performance within the five-country training domain while reducing spatial leakage. Five-fold grouped cross-validation was used. Complete `site_id` groups, rather than individual pixels, were assigned to each validation fold. Therefore, pixels from one mining district or negative sampling site could not occur in both the training and validation data.

A further 10 km exclusion zone was then placed around every validation site. Any candidate training site located inside this distance was removed from that fold. This step was important because nearby pixels can share environmental and spatial characteristics even when they belong to different polygons. Random pixel division could consequently measure local spatial similarity rather than the ability to identify mining in a new location (Roberts et al., 2017; Ploton et al., 2020). Each retained observation received one out-of-fold prediction from a model that had not seen its site or the surrounding 10 km area. This evaluation is referred to as training-region spatial validation throughout the dissertation.

4.2.4 External validation against Amazon Mining Watch

After training-region spatial validation, XGBoost predictions were compared with the Amazon Mining Watch 2024 mining-scar product in Bolivia, French Guiana, Suriname and Venezuela.

These four countries were excluded from model training and were covered by both the held-out labels and Amazon Mining Watch. The binary XGBoost predictions, manual labels and Amazon Mining Watch polygons were spatially joined at the sampled pixel locations. Overall model agreement with Amazon Mining Watch, recall among Amazon Mining Watch-positive pixels and the detected fraction within each intersected Amazon Mining Watch polygon were calculated. Cohen's kappa between the manual labels and Amazon Mining Watch was also reported by country.

This assessment supplied an external reference produced independently of the manual labelling used in this research. However, Amazon Mining Watch is itself derived from a remote-sensing model, and the comparison included only polygons intersecting the labelled samples. It was therefore interpreted as external corroboration at sampled locations rather than ground-truth accuracy or a complete assessment of either product.

4.2.5 Cross-country and cross-continent transfer

After spatial validation, each classifier was fitted once using all observations from Brazil, Peru, Colombia, Guyana and Ecuador. The fitted model was then applied without retraining to two geographically independent groups. Cross-country transfer was evaluated in Bolivia, Venezuela, Suriname and French Guiana. Cross-continent transfer was evaluated in Ghana, Indonesia and Myanmar. No labelled observation from these seven countries was used for model fitting, parameter selection or threshold selection.

The AOIs were centred on documented alluvial-mining districts, and the manually drawn test pixels were designed to include mining surfaces and difficult non-mining alternatives. They were not probability samples of their national territories. The resulting scores therefore measure transfer to the selected held-out environments and do not estimate country-wide map accuracy. A complete continental prediction was also outside the scope of this analysis.

4.2.6 Threshold selection and evaluation metrics

The operating threshold was selected once from the pooled out-of-fold predictions produced by training-region spatial validation. Precision and recall were calculated across possible

thresholds, and the threshold that maximised F1 was retained. For the final XGBoost A_complex model, this value was 0.5406. The same threshold was subsequently applied to every cross-country and cross-continent test observation. Thresholds optimised separately using a held-out country's labels were calculated only as diagnostic upper bounds and were not treated as deployable model results.

F1 was the principal classification metric because it combines precision and recall and does not allow the large negative class to dominate the evaluation in the way that overall accuracy can. Precision showed what proportion of predicted mining pixels were labelled as mining, while recall showed what proportion of labelled mining pixels were detected. Precision-recall area under the curve (PR-AUC) was also reported because it assesses ranking performance across thresholds and is more informative than accuracy for an imbalanced binary problem.

F1 is affected by the proportion of positive observations in an evaluation set. The manually selected country samples contained different positive-to-negative ratios, so directly comparing their raw F1 values would partly compare sampling composition rather than classifier behaviour. Test observations were therefore subsampled within each country to match the 0.414 positive prevalence of the spatial-validation sample. Raw and prevalence-matched results were both retained. Uncertainty around F1 was estimated using 200 non-parametric bootstrap resamples of each evaluation set.

4.2.7 Feature-space analysis

Principal component analysis and UMAP were used to inspect the structure of the AlphaEarth embeddings. PCA measured how much of the 64-dimensional variance could be represented by progressively smaller linear spaces, while UMAP supplied a two-dimensional non-linear visualisation. A five-nearest-neighbour classifier was evaluated in the full embedding and reduced spaces. This analysis tested whether mining and non-mining observations were locally separable within AlphaEarth; it did not replace the geographically independent evaluation.

4.3 Downstream river-change analysis

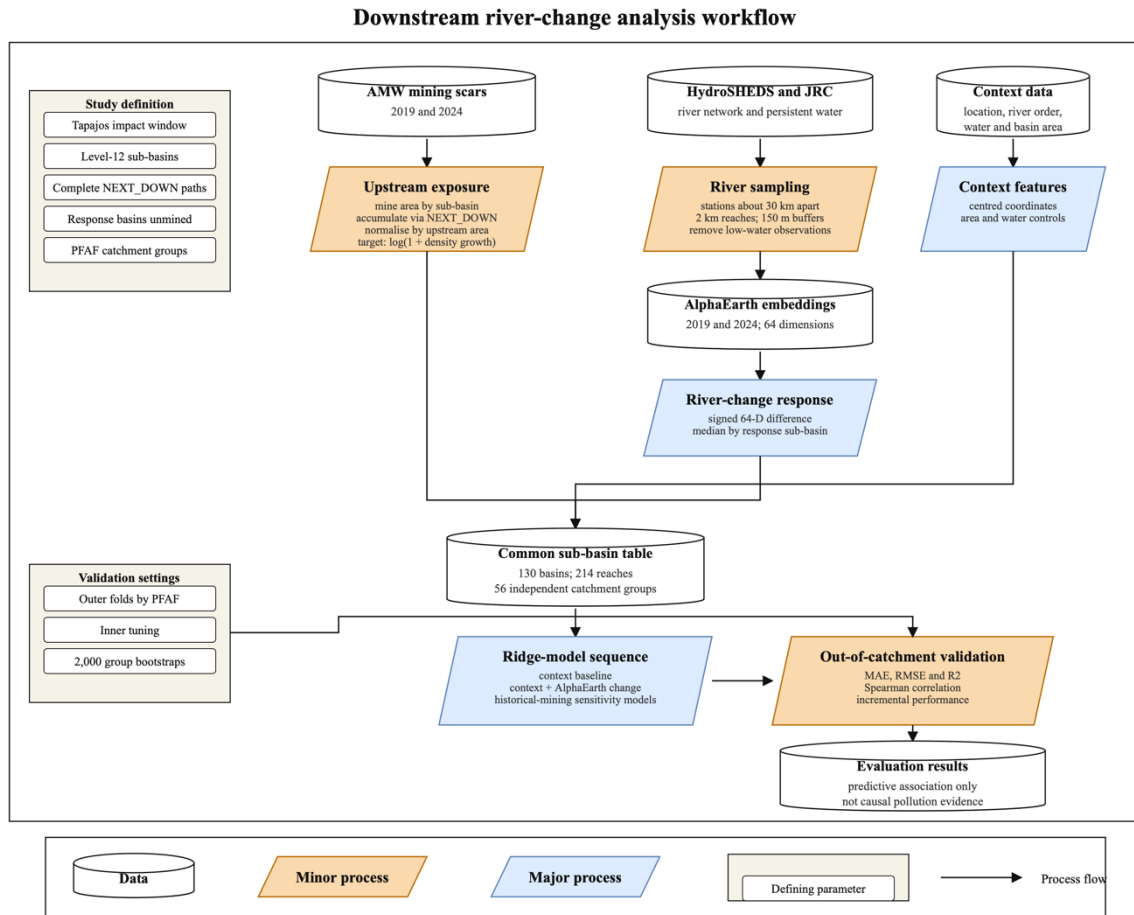


Figure 4.2 Workflow used to test whether downstream AlphaEarth change contains information about hydrologically accumulated upstream mining expansion.

4.3.1 Hydrological exposure to upstream mining

The second analysis used a bounded northern section of the Tapajós river system. The manually defined impact window limited remote-sensing extraction, while the complete connected HydroBASINS topology was retained for calculating upstream exposure. This distinction allowed mines outside the observation window but upstream of a sampled river reach to contribute to its exposure.

Amazon Mining Watch records the first period in which each 10 m mining-scar pixel was detected. Pixels first detected between 2018 and each analysis year were counted within every

HydroBASINS level-12 sub-basin. Mining area was therefore calculated separately for 2019 and 2024. Each sub-basin's local mining area was passed downstream through the NEXT_DOWN field until the outlet of the connected system. The accumulated area was divided by upstream catchment area and expressed as hectares of mining per 1,000 km². The exposure variable was the non-negative increase in this upstream mining density between 2019 and 2024.

This procedure used hydrological connection rather than straight-line distance. A nearby mine did not affect a river observation unless its sub-basin drained towards that location. Sub-basins containing local mining scars in 2024 were excluded from the final river analysis. Consequently, AlphaEarth could not predict mining growth simply by observing the mine surface inside the response unit.

4.3.2 Measuring downstream river change

The HydroSHEDS Free-Flowing Rivers network was clipped to the impact window, and river reaches with RIV_ORD values up to five were retained. Stations were placed at approximately 30 km intervals. Each station represented a 2 km river reach buffered by 150 m on both sides. AlphaEarth values were averaged within this corridor after masking to pixels with at least 30% occurrence in the JRC Global Surface Water record. The station layout was kept identical in 2019 and 2024.

For each paired station, the 2019 value of every AlphaEarth dimension was subtracted from its 2024 value. This produced a signed 64-dimensional change vector. Euclidean magnitude and cosine distance were also calculated, but these unsigned measures were retained only as diagnostics because any environmental change can increase them. Stations with less than 30% valid persistent-water coverage were removed.

Several stations could occur within one level-12 sub-basin, but they shared the same upstream mining exposure. Treating them as independent observations would artificially increase the sample size. The median embedding change and river-context variables were therefore

calculated for each sub-basin. The level-12 sub-basin became the unit of analysis, and the target variable was $\log(1 + \text{growth in upstream mining density})$.

4.3.3 Predictive models and spatial validation

Ridge regression was used to test whether downstream embedding change contained information about upstream mining growth. All predictors were standardised within a scikit-learn pipeline. The models were compared in a fixed sequence. Geography-only predictors contained centred longitude and latitude, their squared terms and their interaction. River-context predictors contained river order, water fraction, log upstream area and log sub-basin area. The non-mining context baseline combined these two groups. The principal full model added all 64 signed AlphaEarth changes to this baseline.

Two sensitivity models then introduced historical mining. The first added log-transformed 2019 upstream mining density to the context baseline. The second added both historical mining and AlphaEarth change. This distinction was required because historical mining was expected to be a strong predictor of later expansion. The principal test asked whether AlphaEarth added information beyond non-mining geographical and river context. The stricter sensitivity test asked whether it retained additional information when prior mining was already known.

Spatial validation grouped level-12 sub-basins by the first eight digits of their PFAF_ID. Five-fold GroupKFold cross-validation held out complete catchment groups, preventing connected neighbouring sub-basins from being divided between training and validation data. Ridge regularisation was selected independently inside each outer training fold using an inner grouped cross-validation across 16 values from 0.001 to 10,000. All reported predictions were therefore out-of-catchment predictions.

Mean absolute error was the principal metric because it reports the typical prediction error without giving exceptional observations the squared influence used by RMSE. RMSE, R^2 and Spearman's rank correlation were also reported. The incremental value of AlphaEarth was measured as the change in MAE and R^2 between the context baseline and the principal full

model. Ninety-five per cent confidence intervals were produced from 2,000 bootstrap resamples of complete PFAF catchment groups rather than individual sub-basins.

4.4 Interpretation and reproducibility boundaries

The two analyses estimate different quantities. The classification analysis measures transfer to labelled pixels in selected mining environments; it does not measure total national mining area or continental map accuracy. The river analysis measures whether downstream AlphaEarth changes predict hydrologically accumulated upstream mining growth after controlling for observable context. It does not demonstrate that mining caused a particular chemical pollutant or sediment concentration.

All intermediate tables retained coordinates, polygon identifiers, site identifiers, country allocation and fold assignment. Earth Engine exports were divided by country or into five deterministic chunks to remain within task limits. These records allow the sample construction, geographical separation and reported predictions to be reproduced without repeating the highest-cost Earth Engine operations.

5. Results

5.1 AlphaEarth feature structure and internal performance

The principal training dataset contained 35,275 observations, of which 41.4% were positive. The three classifiers were evaluated using five-fold training-region spatial validation. Complete sites were retained within the same fold, while candidate training observations located within 10 km of each validation site were excluded. The evaluation therefore measured performance at spatially separated labelled sites within the five-country training region.

Table 5.1. Classifier performance under training-region spatial validation

Classifier	F1	95% interval	PR-AUC
Logistic regression	0.856	0.851–0.860	0.895

Random forest	0.864	0.860–0.868	0.940
XGBoost	0.886	0.882–0.889	0.950

XGBoost achieved the strongest performance, with an F1 of 0.886 and a PR-AUC of 0.950. Random forest produced an F1 of 0.864, while logistic regression achieved 0.856. XGBoost was therefore selected for the subsequent external and geographical-transfer evaluations.

Performance nevertheless varied among the five training countries. F1 ranged from 0.641 in Colombia, where recall was 0.502, to 0.966 in Brazil. Ecuador, Peru and Guyana achieved F1 values of 0.863, 0.895 and 0.914 respectively. The strong pooled result therefore did not represent uniform performance across all training environments.

5.2 Cross-country and cross-continent transfer

The XGBoost model was applied to all held-out countries using the threshold of 0.5406 fixed during training-region spatial validation. Pooled cross-country F1 was 0.833 (95% interval: 0.826-0.840), 0.053 below the internal spatial result. Recall remained high at 0.931, while precision decreased to 0.753. The model therefore detected most labelled mining pixels in the held-out South American countries but produced additional false positives.

Pooled cross-continent F1 was 0.873 (0.867-0.880), only 0.012 below the internal result and 0.040 above cross-country transfer. Indonesia produced the strongest held-out performance at 0.954, followed by Ghana at 0.891. Myanmar was the weakest at 0.772. Performance therefore did not decrease in a simple sequence with geographical distance from the training countries.

All held-out results used labels that were excluded from model fitting and threshold selection. The comparison consequently tested whether the learned AlphaEarth representation transferred beyond the countries supplying training data. It did not constitute complete national mapping, because evaluation remained restricted to independently labelled polygons rather than every pixel within each country.

Table 5.2. Performance in the seven held-out countries

Evaluation	Country	Pixels	Precision	Recall	F1
Cross-country	Bolivia	2,213	0.723	0.967	0.827
Cross-country	French Guiana	2,179	0.860	0.835	0.847
Cross-country	Suriname	4,713	0.771	0.954	0.853
Cross-country	Venezuela	2,564	0.688	0.952	0.799
Cross-continent	Ghana	3,480	0.961	0.830	0.891
Cross-continent	Indonesia	4,389	0.956	0.953	0.954
Cross-continent	Myanmar	3,717	0.714	0.841	0.772

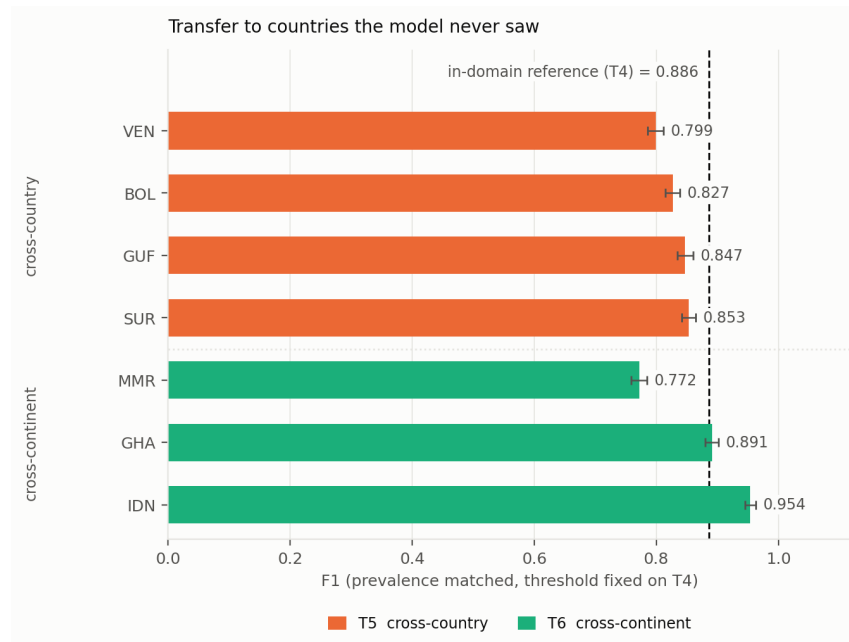


Figure 5.1 Transfer to seven held-out countries. F1 with no pixel from the evaluated country in training. Upper block cross-country (T5), lower cross-continent (T6). Whiskers are bootstrap intervals; the dashed line is the in-domain spatially buffered reference of 0.886. The threshold, fixed at 0.5406 on in-domain folds, was never re-tuned.

The threshold diagnostic showed that PR-AUC remained between 0.934 and 0.982 in six held-out countries. Country-specific threshold optimisation increased F1 by no more than 0.005 in French Guiana, Suriname, Indonesia and Myanmar, although larger gains were possible in Bolivia, Venezuela and Ghana. These are diagnostic ceilings because they use test labels.

Myanmar differed because its F1 and PR-AUC were both the lowest, and changing the threshold increased F1 only from 0.772 to 0.778. Among its labelled negatives, turbid or braided rivers had a mean mining probability of 0.635 and 65.8% exceeded the fixed threshold. Urban or bare negatives had a mean probability of 0.382, with 26.2% exceeding the threshold. Myanmar's lower result therefore reflected weaker class separation rather than only threshold calibration.

5.3 Agreement with Amazon Mining Watch

The labelled sample intersected 26 of the 2,128 Amazon Mining Watch polygons in the four held-out South American countries. Across 14,366 sampled pixels, agreement between XGBoost and Amazon Mining Watch was 0.866. The model identified 91.3% of pixels located inside an Amazon Mining Watch polygon. At polygon level, the median detected fraction was 0.965; all 26 polygons had more than half of their sampled pixels detected, and none was entirely missed. The manually drawn labels also showed high agreement with Amazon Mining Watch, with Cohen's kappa ranging from 0.808 to 1.000.

Table 5.3 Agreement with Amazon Mining Watch at sampled locations

Country	Pixels	Model agreement	Recall of AMW pixels	AMW polygons
Bolivia	3,213	0.867	0.903	8
French Guiana	2,929	0.869	0.839	6
Suriname	4,766	0.863	0.954	6
Venezuela	3,458	0.866	0.944	6
Pooled	14,366	0.866	0.913	26

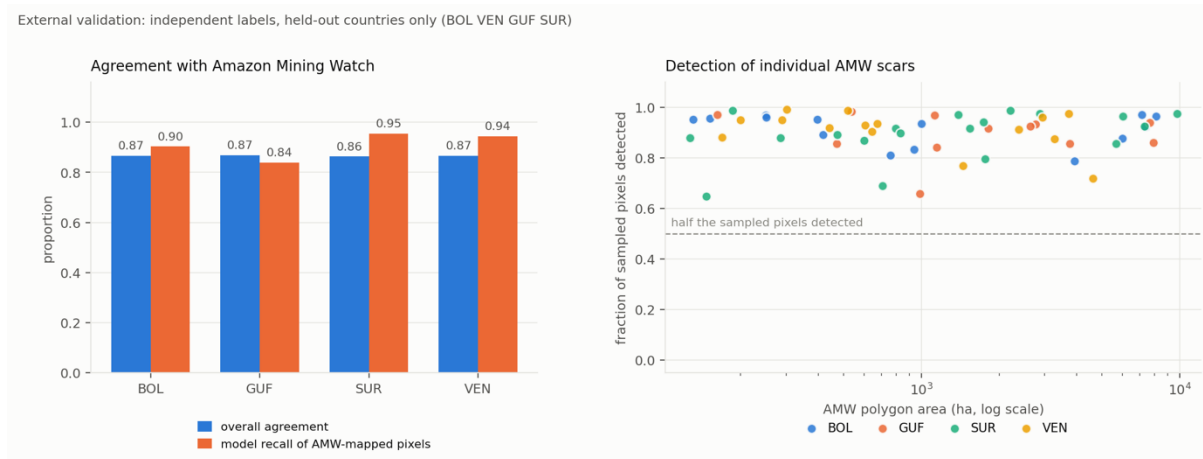


Figure 5.2 Agreement with Amazon Mining Watch, four held-out countries. Left: overall agreement and model recall of AMW-mapped pixels; pooled 0.866 and 0.913. Right: detected fraction per AMW polygon against its area. This is agreement between two automated products, not accuracy, and covers only 26 of 2,128 polygons, all large.

However, the intersected polygons were much larger than the complete Amazon Mining Watch inventory. Their country-level median areas ranged from 376.6 to 3,077.3 ha, compared with an inventory-wide median of 6.48 ha. This result therefore corroborates detection of large mining scars at sampled locations, but does not establish performance for the numerous small scars or provide landscape-level accuracy.

5.4 Upstream mining growth and downstream river observations

The downstream analysis covered 4,170 HydroBASINS level-12 sub-basins. Amazon Mining Watch recorded 157,008 ha of cumulative mining scar by 2019 and 265,274 ha by 2024, an increase of 69.0%. Of 619 river stations with paired AlphaEarth observations, 279 met the permanent-water requirement. After locally mined sub-basins were removed and repeated river reaches were aggregated, 130 sub-basins, 214 contributing reaches and 56 independent PFAF catchment groups remained.

The median increase in upstream mining density was 181.3 ha per 1,000 km², with an interquartile range from 77.9 to 225.2. The 90th percentile was 575.5 and the maximum was

1,614.1 ha per 1,000 km². The analysis therefore retained a continuous exposure gradient rather than dividing the observations into high- and low-mining groups.

5.5 Predicting upstream mining growth from downstream change

The model using geographical and river context achieved an out-of-catchment R^2 of 0.412 and MAE of 1.127. AlphaEarth change alone was weaker, with R^2 of 0.162 and MAE of 1.310. However, combining context with the 64 signed AlphaEarth changes increased R^2 to 0.626 and reduced MAE to 0.863. Its Spearman correlation between observed and predicted growth was 0.760 ($p < 0.001$).

Table 5.4 Out-of-catchment prediction of upstream mining growth

Predictor set	Features	MAE	RMSE	R^2	Spearman rho
Geography and river context	9	1.1267	1.4436	0.4115	0.7068
Unsigned AlphaEarth magnitude	2	1.5432	1.9874	-0.1154	-0.1054
Signed AlphaEarth change	64	1.3096	1.7223	0.1623	0.5495
Context and signed AlphaEarth change	73	0.8632	1.1503	0.6263	0.7596
Context and historical mining	10	0.1206	0.1878	0.9900	0.9640
Context, historical mining and AlphaEarth change	74	0.1910	0.2646	0.9802	0.9248

Relative to the context baseline, AlphaEarth reduced MAE by a bootstrap mean of 0.2609 (95% catchment interval: 0.0864-0.4876) and increased R^2 by 0.2005 (0.0265-0.3721). Both intervals remained above zero. AlphaEarth change therefore added predictive information about upstream mining growth beyond non-mining geographical and river context.

Because entire PFAF catchment groups were held out together, this improvement was measured on river systems not used to fit the model. The result was therefore not driven solely by repeated stations from the same connected network. Nevertheless, the sample contained only 56

independent catchment groups, and the response represented mapped mining-scar expansion rather than field-measured contaminant concentration.

Historical mining was much stronger. Supplying 2019 upstream mining density produced R^2 of 0.990 and MAE of 0.121. Adding AlphaEarth then reduced performance: the bootstrap mean change in MAE was -0.0705 (-0.1042 to -0.0389), where a negative value indicates increased error. AlphaEarth was therefore informative when mining history was unavailable, but added no value when the earlier mining inventory was already known.

Unsigned embedding magnitude produced a negative R^2 of -0.115 and a non-significant Spearman correlation of -0.105 ($p = 0.233$). The observed signal depended on the direction of change across the 64 dimensions rather than the total amount of embedding movement.

The planned NDTI comparison did not produce a result because paired 2019 and 2024 basin-level exports were unavailable to the final notebook. Therefore, the downstream result demonstrates a predictive association with upstream mining growth, not a direct measurement of suspended sediment or evidence of causality.

6. Discussion

6.1 Cross-country and cross-continent transfer

The first objective examined whether a classifier trained from geographically limited labels could identify alluvial mining in unseen countries. The results provide qualified support for this objective. XGBoost achieved an F1 of 0.886 under training-region spatial validation, 0.833 across four held-out South American countries and 0.873 across Ghana, Indonesia and Myanmar. All transfer results used the threshold of 0.5406 selected inside the training data, without country-specific retraining. Therefore, the held-out scores represent genuine transfer to the selected test environments rather than adaptation to their labels.

The cross-country result was characterised by high recall of 0.931 and lower precision of 0.753. This balance is useful for initial screening because relatively few labelled mining pixels were missed, but it also means that predicted candidates cannot be treated as confirmed mines. XGBoost exceeded random forest and logistic regression under the same spatial validation, suggesting that non-linear interactions among the 64 embedding dimensions were useful. However, the difference between classifiers was smaller than the variation among countries. Data coverage and environmental representativeness therefore remained at least as important as model choice.

The relatively small decline between internal spatial validation and cross-continent testing is important. Houriez et al. (2025) similarly used simple classifiers to transfer AlphaEarth vegetation labels from the United States to Canada, obtaining 81% and 73% accuracy on the same 13-class task. The present result extends this transfer proposition to alluvial mining. However, Indonesia achieved an F1 of 0.954, Myanmar achieved 0.772, and Colombia was weaker than both within the training domain. Performance therefore did not follow geographical distance. Local mine morphology, water conditions, vegetation and negative-label composition remained more important.

Myanmar reveals the main failure mechanism. Turbid and braided rivers received a mean mining probability of 0.635, and 65.8% exceeded the fixed threshold. Threshold adjustment produced little improvement because PR-AUC was also lowest, indicating weak class separation rather than only poor calibration. Geographical changes in image and class distributions are a recognised obstacle to remote-sensing transfer (Tuia, Persello and Bruzzone, 2016). Ade, Ghorbanzadeh and Blaschke (2026) similarly found that training geography affected mining-model performance more than architecture or radar inclusion. Here, XGBoost did not explicitly represent the geometric contrast between irregular mine components and a connected linear river. It transferred surface characteristics effectively, but not landscape geometry.

The seven test datasets were designed around relevant mining environments and difficult negative classes. They are independent test locations, but not probability samples of each

country. The results should therefore be described as cross-country and cross-continent transfer across selected alluvial-mining environments, rather than continental mapping accuracy. This distinction also explains why the higher cross-continent score does not establish that intercontinental transfer is easier than transfer within South America. It reflects the particular labelled environments included in each group.

6.2 Upstream mining and downstream AlphaEarth change

The second objective asked whether downstream AlphaEarth change contained information about hydrologically connected upstream mining growth. Between 2019 and 2024, mapped mining scar increased by 69.0% across the study system. After water masking, removal of locally mined sub-basins and spatial aggregation, the analysis retained 130 downstream sub-basins within 56 independent catchment groups. This design prevented the model from reading a mine located inside the response unit and required it to predict upstream growth from downstream conditions.

River and geographical context alone produced an out-of-catchment R^2 of 0.412 and an MAE of 1.127. Adding the 64 signed AlphaEarth changes increased R^2 to 0.626 and reduced MAE to 0.863. The bootstrap intervals for both improvements remained above zero, while the correlation between observed and predicted growth reached 0.760. AlphaEarth therefore contained information associated with upstream mining expansion that was not explained by river order, water coverage, catchment area or geographical location alone. Because complete catchment groups were held out during validation, the improvement cannot be attributed only to neighbouring stations from the same river network.

The form of the signal is also informative. Unsigned embedding magnitude produced negative R^2 and no significant rank correlation, while the signed 64-dimensional change was predictive. A general amount of environmental change was insufficient; the direction of change within the embedding space mattered. This is consistent with AlphaEarth representing several dimensions of annual surface condition rather than one interpretable water-quality index. However, individual embedding axes do not correspond directly to suspended sediment, mercury or any

other pollutant. The result demonstrates an indirect monitoring signal, not a physical measurement of pollution.

Historical mining provides a second boundary. Context and 2019 mining density predicted subsequent growth with an R^2 of 0.990, and AlphaEarth did not improve this model. Mining expansion was highly path-dependent: locations already affected by mining were the strongest predictors of further expansion. AlphaEarth is therefore most valuable when a reliable historical mining inventory is unavailable or outdated. It is not a replacement for an accurate existing time series.

These findings are compatible with independent evidence that mining can alter downstream sediment transport. In French Guiana, Gallay et al. (2018) combined monthly field suspended-sediment measurements, MODIS water-colour observations, land-use change and hydroclimatic analysis to identify mining-related changes in sediment fluxes. It can therefore be concluded that downstream embeddings helped predict upstream mapped mining growth, but not that the observed embedding changes quantified pollution or established causality.

6.3 Value of the research

This research shows a practical use of a newly released foundation-model dataset for a monitoring problem where labels, computing capacity and field access are limited. Instead of downloading and compositing large archives of optical and radar imagery, 64 analysis-ready variables were sampled only at labelled pixels and selected river reaches. This allowed model development with Google Earth Engine and free Google Colab resources while retaining 10 m information. The design is relevant to agencies and researchers that cannot maintain a bespoke deep-learning and image-processing infrastructure.

The contribution is evidence that limited labels can support geographically separated screening, followed by a second demonstration that annual embeddings can contain information connected to change elsewhere in a river network. These results position AlphaEarth as a prioritisation tool: it can identify places requiring image interpretation, higher-resolution

mapping or field inspection. Human verification remains necessary, especially where turbid rivers, agriculture and exposed ground resemble mining.

6.4 Overall methodological design

The methodological design was appropriate for testing transferability under limited data and computing resources, but it can be improved in several ways. First, higher-resolution satellite, aerial or drone imagery could support more precise labels and resolve narrow riverbanks, small excavations and mixed pixels that remained difficult at 10 m. Independent field observations and labels produced by multiple interpreters would also provide stronger validation.

Second, greater computing capacity would allow complete pixel-level inference across larger regions, denser sampling and repeated analysis across additional years. This would convert the present evaluation at labelled locations into a more complete assessment of regional coverage. It would also permit wider parameter testing without reducing the spatial extent of the analysis.

Third, alternative algorithms could represent the context that the pixel-based XGBoost model could not read directly. Convolutional neural networks could examine surrounding image patches and learn mine shape and texture, while graph neural networks could represent connections among river reaches, sub-basins and upstream mining areas. A hybrid workflow could retain AlphaEarth for low-cost candidate screening and apply contextual models only to selected locations. Future analysis should also include probability-based test sampling, independent mining inventories and physical measurements of suspended sediment or water quality.

6.5 Comparison with previous studies

Direct numerical comparison requires caution because studies use different mine definitions, label prevalence, spatial partitions and metrics. Ade, Ghorbanzadeh and Blaschke (2026) trained Sentinel-1 and Sentinel-2 segmentation models in Ghana and Côte d'Ivoire. Their strongest model achieved an F1 of 0.856 within Ghana, while transfer from Ghana to Côte d'Ivoire produced 0.656; combined-country training increased it to 0.724. This confirms that

transfer can be asymmetric and benefits from geographical label diversity. The present F1 scores of 0.833 and 0.873 concern labelled pixels rather than exhaustive segmentation and are not directly superior.

Houriez et al. (2025) provide an independent comparison for AlphaEarth rather than mining. Simple classifiers transferred a 13-class vegetation dataset from the United States to Canada, with validation accuracy declining from 81% to 73%. Both studies show that lightweight classifiers can operate beyond the geography of their labels. This research adds country- and continent-held-out alluvial-mining tests, although the preprint's land-cover task is not a mining benchmark.

Forkuor, Ullmann and Griesbeck (2020) used Sentinel-1 time series to identify annual mining changes in south-western Ghana without cloud-free optical scenes. However, approximately 25% of radar observations were unusable because of high-intensity rainfall effects. AlphaEarth avoids constructing a bespoke sensor time series, but its annual pixel representation contains less explicit temporal and geometric information. Radar time series, neighbourhood texture or object-shape variables may therefore reduce river-shaped false positives where embeddings alone do not separate classes.

6.6 Future work

The immediate priority is broader and more independent validation. Future labels should be selected through a documented sampling frame stratified by country, mine size, river setting and known hard negatives. Independent interpreters, field observations and government inventories should be used where possible. Evaluation should deliberately include small and partially vegetated scars to determine whether the current result extends beyond large, clearly visible mining complexes.

The classifier should next incorporate context without abandoning the low-cost design. Candidate approaches include neighbourhood summaries of AlphaEarth, connected-component shape filters, or a lightweight CNN applied only to high-probability candidate

windows. A coarse-to-fine workflow could first screen large areas at lower computational cost and then evaluate candidate locations at 10 m. This would convert transfer performance into map coverage while limiting Earth Engine consumption.

For downstream monitoring, the analysis should be repeated across additional river systems and years. Paired optical indices, satellite-derived suspended sediment and field measurements would determine what part of the embedding signal represents water quality. Models should also control for rainfall, discharge, dams, deforestation and fire. A prospective test, in which earlier embeddings predict mining expansion recorded only in a later inventory, would provide stronger temporal evidence. Until these tests are completed, AlphaEarth should be presented as a transferable and computationally efficient screening representation, not as an autonomous mine map or a direct pollution sensor.

7. Conclusion

This dissertation evaluated whether AlphaEarth can support efficient alluvial-mining monitoring across borders. The findings show that models trained with limited AlphaEarth samples can identify mining in unseen countries on different continents without local retraining. However, performance varied among environments, particularly where turbid rivers, exposed ground and local mine morphology resembled labelled mining surfaces.

The study found that changes in downstream AlphaEarth embeddings contained information associated with hydrologically connected upstream mining expansion beyond geographical and river context. This predictive advantage disappeared when an accurate historical mining inventory was available. AlphaEarth should therefore be understood as a low-processing screening representation where labels, computing capacity and inventories are limited. It can prioritise locations for mapping and inspection, but cannot replace independent validation, regional mapping or water-quality measurements, nor does it establish pollution causality.

Code Availability: <https://github.com/Entao21>

References

- Ade, B., Ghorbanzadeh, O. and Blaschke, T. (2026) 'Cross-country transferability of deep learning models for artisanal and small-scale mining mapping in Ghana and Côte d'Ivoire', *Geosciences*, 16(8), 310. <https://doi.org/10.3390/geosciences16080310>.
- Agyapong, D., Cammeraat, E., Jansen, B. and Krueger, E.H. (2026) 'Artisanal and small-scale gold mining effects in social-ecological systems: A systematic literature review using the DPSIR framework', *The Extractive Industries and Society*, 26, 101879. doi:10.1016/j.exis.2026.101879.
- Alessi, M.A., Chirico, P.G. and Millones, M. (2023) 'Artisanal mining river dredge detection using SAR: A method comparison', *Remote Sensing*, 15(24), article 5701. Available at: <https://doi.org/10.3390/rs15245701>.
- Alvarez-Berrios, N.L. and Aide, T.M. (2015) 'Global demand for gold is another threat for tropical forests', *Environmental Research Letters*, 10(1), 014006. Available at: <https://doi.org/10.1088/1748-9326/10/1/014006>.
- Asner, G.P., Llactayo, W., Tupayachi, R. and Luna, E.R. (2013) 'Elevated rates of gold mining in the Amazon revealed through high-resolution monitoring', *Proceedings of the National Academy of Sciences*, 110(46), pp. 18454-18459. Available at: <https://doi.org/10.1073/pnas.1318271110>.
- Balaniuk, R., Isupova, O. and Reece, S. (2020) 'Mining and tailings dam detection in satellite imagery using deep learning', *Sensors*, 20(23), 6936. Available at: <https://doi.org/10.3390/s20236936>.
- Baloyi, J., Ramdhani, N., Mbhele, R. and Simate, G.S. (2024) 'Acid mine drainage from gold mining in South Africa: remediation, reuse, and resource recovery', *Mine Water and the Environment*, 43, pp. 418–430. Available at: <https://doi.org/10.1007/s10230-024-00994-2>.
- Becerra, M., Villa, L., Nicolau, A.P. et al. (2024) 'Creating near real-time alerts of illegal gold mining in the Peruvian Amazon using synthetic aperture radar', *Environmental Research Communications*, 6(12), 125022. Available at: <https://doi.org/10.1088/2515-7620/ad937e>.

- Belgiu, M. and Dragut, L. (2016) 'Random forest in remote sensing: a review of applications and future directions', *ISPRS Journal of Photogrammetry and Remote Sensing*, 114, pp. 24-31. Available at: <https://doi.org/10.1016/j.isprsjprs.2016.01.011>.
- Boamah, S.P. (2016) 'Alluvial gold mining and our muddy waters', *Modern Ghana*, 1 June. Available at: <https://www.modernghana.com/news/695508/alluvial-gold-mining-and-our-muddy-waters.html>
- Brown, C.F., Kazmierski, M.R., Pasquarella, V.J. et al. (2025) 'AlphaEarth Foundations: an embedding field model for accurate and efficient global mapping from sparse label data', *arXiv*. Available at: <https://doi.org/10.48550/arXiv.2507.22291>.
- Caballero Espejo, J., Messinger, M., Roman-Danobeytia, F., Ascorra, C., Fernandez, L.E. and Silman, M. (2018) 'Deforestation and forest degradation due to gold mining in the Peruvian Amazon: a 34-year perspective', *Remote Sensing*, 10(12), 1903. Available at: <https://doi.org/10.3390/rs10121903>.
- Camalan, S., Cui, K., Pauca, V.P. et al. (2022) 'Change detection of Amazonian alluvial gold mining using deep learning and Sentinel-2 imagery', *Remote Sensing*, 14(7), 1746. Available at: <https://doi.org/10.3390/rs14071746>.
- Chen, T. and Guestrin, C. (2016) 'XGBoost: a scalable tree boosting system', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York: ACM, pp. 785-794. Available at: <https://doi.org/10.1145/2939672.2939785>.
- Couttenier, M., Di Rollo, S., Inguere, L., Mohand, M. and Schmidt, L. (2022) 'Mapping artisanal and small-scale mines at large scale from space with deep learning', *PLOS ONE*, 17(9), e0267963. Available at: <https://doi.org/10.1371/journal.pone.0267963>.
- Couttenier, M., Di Rollo, S., Inguere, L., Mohand, M. and Schmidt, L. (2022) 'Mapping artisanal and small-scale mines at large scale from space with deep learning', *PLOS ONE*, 17(9), e0267963. Available at: <https://doi.org/10.1371/journal.pone.0267963>.
- Dethier, E.N., Renshaw, C.E. and Magilligan, F.J. (2022) 'Rapid changes to global river suspended sediment flux by humans', *Science*, 376(6600), pp. 1447-1452. Available at: <https://doi.org/10.1126/science.abn7980>.
- Dethier, E.N., Sartain, S.L. and Lutz, D.A. (2019) 'Heightened levels and seasonal inversion of riverine suspended sediment in a tropical biodiversity hot spot due to artisanal gold

mining', *Proceedings of the National Academy of Sciences*, 116(48), pp. 23936-23941. Available at: <https://doi.org/10.1073/pnas.1907842116>.

Dethier, E.N., Silman, M., Leiva, J.D. et al. (2023) 'A global rise in alluvial mining increases sediment load in tropical rivers', *Nature*, 620(7975), pp. 787-793. Available at: <https://doi.org/10.1038/s41586-023-06309-9>.

Diringer, S.E., Berky, A.J., Marani, M. et al. (2020) 'Deforestation due to artisanal and small-scale gold mining exacerbates soil and mercury mobilization in Madre de Dios, Peru', *Environmental Science & Technology*, 54(1), pp. 286-296. Available at: <https://doi.org/10.1021/acs.est.9b06620>.

Diringer, S.E., Feingold, B.J., Ortiz, E.J. et al. (2015) 'River transport of mercury from artisanal and small-scale gold mining and risks for dietary mercury exposure in Madre de Dios, Peru', *Environmental Science: Processes & Impacts*, 17(2), pp. 478-487. Available at: <https://doi.org/10.1039/C4EM00567H>.

Dogliotti, A.I., Ruddick, K.G., Nechad, B., Doxaran, D. and Knaeps, E. (2015) 'A single algorithm to retrieve turbidity from remotely-sensed data in all coastal and estuarine waters', *Remote Sensing of Environment*, 156, pp. 157-168. Available at: <https://doi.org/10.1016/j.rse.2014.09.020>.

EITI (2023) EITI Standard 2023. Oslo: Extractive Industries Transparency Initiative. Available at: <https://eiti.org/eiti-standard> (Accessed: 4 September 2026).

Elmes, A., Ipanaque, J.G.Y., Rogan, J. et al. (2014) 'Mapping licit and illicit mining activity in the Madre de Dios region of Peru', *Remote Sensing Letters*, 5(10), pp. 882-891. Available at: <https://doi.org/10.1080/2150704X.2014.973080>.

Ericsson, M. and Löf, O. (2019) 'Mining's contribution to national economies between 1996 and 2016', *Mineral Economics*, 32, pp. 223-250. Available at: <https://doi.org/10.1007/s13563-019-00191-6>.

Esdaile, L.J. and Chalker, J.M. (2018) 'The mercury problem in artisanal and small-scale gold mining', *Chemistry - A European Journal*, 24(27), pp. 6905-6916. Available at: <https://doi.org/10.1002/chem.201704840>.

Esdaile, L.J. and Chalker, J.M. (2018) 'The mercury problem in artisanal and small-scale gold mining', *Chemistry – A European Journal*, 24(27), pp. 6905-6916. Available at: <https://doi.org/10.1002/chem.201704840>.

- Espinoza-Villar, R., Martinez, J.M., Armijos, E., Espinoza, J.C., Filizola, N., Dos Santos, A., Willems, B., Fraizy, P., Santini, W. and Vauchel, P. (2018) 'Spatio-temporal monitoring of suspended sediments in the Solimões River (2000–2014)', *Comptes Rendus Geoscience*, 350(1–2), pp. 4–12. Available at: <https://doi.org/10.1016/j.crte.2017.05.001>.
- Ferreira Neto, J.A. et al. (2024) 'Uncontrolled illegal mining and garimpo in the Brazilian Amazon', *Nature Communications*, 15, 9847. Available at: <https://doi.org/10.1038/s41467-024-54220-2>.
- Fonseca, A., Marshall, M.T. and Salama, S. (2024) 'Enhanced detection of artisanal small-scale mining with spectral and textural segmentation of Landsat time series', *Remote Sensing*, 16(10), 1749. Available at: <https://doi.org/10.3390/rs16101749>.
- Forkuor, G., Ullmann, T. and Griesbeck, M. (2020) 'Mapping and monitoring small-scale mining activities in Ghana using Sentinel-1 time series (2015-2019)', *Remote Sensing*, 12(6), 911. <https://doi.org/10.3390/rs12060911>.
- Fritz, M., McQuilken, J., Collins, N. and Weldegiorgis, F. (2018) *Global Trends in Artisanal and Small-Scale Mining: A Review of Key Numbers and Issues*. Winnipeg: International Institute for Sustainable Development. Available at: <https://www.iisd.org/system/files/publications/igf-asm-global-trends.pdf>.
- Fritz, M., McQuilken, J., Collins, N. and Weldegiorgis, F. (2018) *Global Trends in Artisanal and Small-Scale Mining: A Review of Key Numbers and Issues*. Winnipeg: International Institute for Sustainable Development. Available at: <https://www.iisd.org/system/files/publications/igf-asm-global-trends.pdf> (Accessed: 4 September 2026).
- Gallay, M., Martinez, J.-M., Allo, S., Mora, A., Cochonneau, G., Gardel, A., Doudou, J.-C., Sarrazin, M., Chow-Toun, F. and Laraque, A. (2018) 'Impact of land degradation from mining activities on the sediment fluxes in two large rivers of French Guiana', *Land Degradation & Development*, 29(12), pp. 4323-4336. <https://doi.org/10.1002/ldr.3150>.
- Gallwey, J., Robiati, C., Coggan, J., Vogt, D. and Eyre, M. (2020) 'A Sentinel-2 based multispectral convolutional neural network for detecting artisanal small-scale mining in Ghana: applying deep learning to shallow mining', *Remote Sensing of Environment*, 248, 111970. Available at: <https://doi.org/10.1016/j.rse.2020.111970>.

- Gerson, J.R., Szponar, N., Almeyda Zambrano, A. et al. (2022) 'Amazon forests capture high levels of atmospheric mercury pollution from artisanal gold mining', *Nature Communications*, 13, 559. Available at: <https://doi.org/10.1038/s41467-022-27997-3>.
- Gibb, H. and O'Leary, K.G. (2014) 'Mercury exposure and health impacts among individuals in the artisanal and small-scale gold mining community: a comprehensive review', *Environmental Health Perspectives*, 122(7), pp. 667-672. Available at: <https://doi.org/10.1289/ehp.1307864>.
- Global Reporting Initiative (GRI) (2024) GRI 14: Mining Sector 2024. Amsterdam: Global Reporting Initiative. Available at: <https://www.globalreporting.org/pdf.ashx?id=12024>.
- Google Earth Engine and Google DeepMind (2025) Satellite Embedding V1 Annual [dataset]. Available at: https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_SATELLITE_EMBEDDING_V1_ANNUAL.
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. and Moore, R. (2017) 'Google Earth Engine: planetary-scale geospatial analysis for everyone', *Remote Sensing of Environment*, 202, pp. 18-27. Available at: <https://doi.org/10.1016/j.rse.2017.06.031>.
- Hentschel, T., Hruschka, F. and Priester, M. (2002) *Global Report on Artisanal & Small-Scale Mining*. MMSD Working Paper No. 70. London: International Institute for Environment and Development. Available at: <https://www.iied.org/g00723>.
- Hilson, G. (2016) 'Farming, small-scale mining and rural livelihoods in Sub-Saharan Africa: A critical overview', *The Extractive Industries and Society*, 3(2), pp. 547–563. doi:10.1016/j.exis.2016.02.003.
- Hirons, M. (2020) 'How the Sustainable Development Goals risk undermining efforts to address environmental and social issues in the small-scale mining sector', *Environmental Science & Policy*, 114, pp. 321-328. Available at: <https://doi.org/10.1016/j.envsci.2020.08.022>.
- Houriez, L., Pilarski, S., Vahedi, B., Ahmadalipour, A., Honda Scully, T., Aflitto, N., Andre, D., Jaffe, C., Wedner, M., Mazzola, R., Jeffery, J., Messinger, B., McGinley-Smith, S. and Russell, S. (2025) 'Scalable geospatial data generation using AlphaEarth Foundations model', arXiv:2508.11739. <https://doi.org/10.48550/arXiv.2508.11739>.

- Ibrahim, E., Lema, L., Barnabé, P., Lacroix, P. and Pirard, E. (2020) 'Small-scale surface mining of gold placers: Detection, mapping, and temporal analysis through the use of free satellite imagery', *International Journal of Applied Earth Observation and Geoinformation*, 93, article 102194. Available at: <https://doi.org/10.1016/j.jag.2020.102194>.
- IFC (2012) Performance Standards on Environmental and Social Sustainability. Washington, DC: International Finance Corporation. Available at: <https://www.ifc.org/content/dam/ifc/doc/2023/ifc-performance-standards-2012-en.pdf>.
- IGF (2023) IGF Mining Policy Framework. Winnipeg: Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development. Available at: <https://www.igfmining.org/resource/igf-mining-policy-framework/>.
- ILO (1999) Social and Labour Issues in Small-Scale Mines. Report TMSSM/1999. Geneva: International Labour Office. Available at: <https://www.ilo.org/resource/social-and-labour-issues-small-scale-mines-report-tmssm1999>.
- Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF) (2023) IGF Mining Policy Framework. Winnipeg: International Institute for Sustainable Development. Available at: <https://www.igfmining.org/resource/igf-mining-policy-framework/>.
- International Atomic Energy Agency (IAEA) (2025) Monitoring for Protection of the Public and the Environment. IAEA Safety Standards Series No. GSG-19. Vienna: International Atomic Energy Agency. Available at: <https://doi.org/10.61092/iaea.i7sy-xqdl>.
- International Finance Corporation (IFC) (2012) Performance Standards on Environmental and Social Sustainability. Washington, DC: International Finance Corporation. Available at: <https://www.ifc.org/content/dam/ifc/doc/2023/ifc-performance-standards-2012-en.pdf>.
- Isidro, C.M., McIntyre, N., Lechner, A.M. and Callow, I. (2017) 'Applicability of Earth Observation for identifying small-scale mining footprints in a wet tropical region', *Remote Sensing*, 9(9), 945. Available at: <https://doi.org/10.3390/rs9090945>.
- Jiang, D., Matsushita, B., Pahlevan, N. et al. (2023) 'Estimating the concentration of total suspended solids in inland and coastal waters from Sentinel-2 MSI: a semi-analytical approach', *ISPRS Journal of Photogrammetry and Remote Sensing*, 204, pp. 362-377. Available at: <https://doi.org/10.1016/j.isprsjprs.2023.09.020>.

- Kemp, P., Sear, D., Collins, A., Naden, P. and Jones, I. (2011) 'The impacts of fine sediment on riverine fish', *Hydrological Processes*, 25(11), pp. 1800-1821. Available at: <https://doi.org/10.1002/hyp.7940>.
- Lobo, F.L., Costa, M.P.F. and Novo, E.M.L.M. (2016) 'Distribution of artisanal and small-scale gold mining in the Tapajos River Basin over the past 40 years and relationship with water siltation', *Remote Sensing*, 8(7), 579. Available at: <https://doi.org/10.3390/rs8070579>.
- Lobo, F.L., Souza-Filho, P.W.M., Novo, E.M.L.M., Carlos, F.M. and Barbosa, C.C.F. (2018) 'Mapping mining areas in the Brazilian Amazon using MSI/Sentinel-2 imagery', *Remote Sensing*, 10(8), 1178. Available at: <https://doi.org/10.3390/rs10081178>.
- Loots, M., Grobbelaar, S. and Van der Lingen, E. (2022) 'A review of remote-sensing unmanned aerial vehicles in the mining industry', *Journal of the Southern African Institute of Mining and Metallurgy*, 122(7), pp. 387–396. Available at: <https://doi.org/10.17159/2411-9717/1602/2022>.
- Mathioudakis, S., Xiroudakis, G., Petrakis, E. and Manoutsoglou, E. (2023) 'Alluvial Gold Mining Technologies from Ancient Times to the Present', *Mining*, 3(4), pp. 618–644. doi:10.3390/mining3040034.
- Maus, V., Giljum, S., da Silva, D.M. et al. (2022) 'An update on global mining land use', *Scientific Data*, 9, 433. Available at: <https://doi.org/10.1038/s41597-022-01547-4>.
- Maus, V., Giljum, S., Gutschlhofer, J., da Silva, D.M., Probst, M., Gass, S.L.B., Luckeneder, S., Lieber, M. and McCallum, I. (2020) 'A global-scale data set of mining areas', *Scientific Data*, 7, article 289. Available at: <https://doi.org/10.1038/s41597-020-00624-w>.
- Mestanza-Ramón, C. et al. (2022) 'Gold Mining in the Amazon Region of Ecuador: History and a Review of Its Socio-Environmental Impacts', *Land*, 11(2), 221. doi:10.3390/land11020221.
- Meutia, A.A., Lumowa, R. and Sakakibara, M. (2022) 'Indonesian Artisanal and Small-Scale Gold Mining—A Narrative Literature Review', *International Journal of Environmental Research and Public Health*, 19(7), 3955. doi:10.3390/ijerph19073955.
- Minamata Convention on Mercury (2013) Text and Annexes. Geneva: Secretariat of the Minamata Convention on Mercury. Available at:

<https://minamataconvention.org/en/resources/minamata-convention-mercury-text-and-annexes>.

- Mol, J.H. and Ouboter, P.E. (2004) 'Downstream effects of erosion from small-scale gold mining on the instream habitat and fish community of a small Neotropical rainforest stream', *Conservation Biology*, 18(1), pp. 201–214. Available at: <https://doi.org/10.1111/j.1523-1739.2004.00080.x>.
- Ngom, N.M. et al. (2023) 'Artisanal exploitation of mineral resources: Remote sensing observations of environmental consequences, social and ethical aspects', *Surveys in Geophysics*, 44, pp. 225–247. Available at: <https://doi.org/10.1007/s10712-022-09740-1>.
- Nsiah, R.A., Laari, P.B. and Guan, Q. (2026) 'Mapping mining expansion in the Aprampama Forest Reserve with low-shot learning and satellite embeddings', in FIG Congress 2026: The Future We Want - The SDGs and Beyond, Cape Town, South Africa, 24-29 May. Available at: https://www.fig.net/resources/proceedings/fig_proceedings/fig2026/papers/ts11h/TS11H_nsiah_laari_et_al_13647.pdf (Accessed: 4 September 2026).
- Nursamsi, I., Phinn, S.R., Levin, N., Luskin, M.S. and Sonter, L.J. (2024) 'Remote sensing of artisanal and small-scale mining: a review of scalable mapping approaches', *Science of the Total Environment*, 951, article 175761. Available at: <https://doi.org/10.1016/j.scitotenv.2024.175761>.
- Nursamsi, I., Phinn, S.R., Levin, N., Luskin, M.S. and Sonter, L.J. (2024) 'Remote sensing of artisanal and small-scale mining: a review of scalable mapping approaches', *Science of the Total Environment*, 951, 175761. Available at: <https://doi.org/10.1016/j.scitotenv.2024.175761>.
- OECD (2016) *OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas*. 3rd edn. Paris: OECD Publishing. Available at: <https://doi.org/10.1787/9789264252479-en>.
- Ondayo, M.A., Watts, M.J., Mitchell, C.J., King, D.C.P. and Osano, O. (2024) 'Review: Artisanal Gold Mining in Africa—Environmental Pollution and Human Health Implications', *Exposure and Health*, 16, pp. 1067–1095. doi:10.1007/s12403-023-00611-7.
- Park, E. and Latrubesse, E.M. (2014) 'Modeling suspended sediment distribution patterns of the Amazon River using MODIS data', *Remote Sensing of Environment*, 147, pp. 232–242. Available at: <https://doi.org/10.1016/j.rse.2014.03.013>.

- Ploton, P., Mortier, F., Rejou-Mechain, M. et al. (2020) 'Spatial validation reveals poor predictive performance of large-scale ecological mapping models', *Nature Communications*, 11, 4540. Available at: <https://doi.org/10.1038/s41467-020-18321-y>.
- Prescott, G.W., Baird, M., Geenen, S., Nkuba, B., Phelps, J. and Webb, E.L. (2022) 'Formalizing artisanal and small-scale gold mining: a grand challenge of the Minamata Convention', *One Earth*, 5(3), pp. 242-251. Available at: <https://doi.org/10.1016/j.oneear.2022.02.005>.
- Prescott, G.W., Maung, A.C., Aung, Z. et al. (2020) 'Gold, farms, and forests: enforcement and alternative livelihoods are unlikely to disincentivize informal gold mining', *Conservation Science and Practice*, 2(3), e142. Available at: <https://doi.org/10.1111/csp2.142>.
- Ren, H. et al. (2026) 'Global mining has undermined forest conservation within and beyond protected areas', *Nature Communications*, 17, article 7954. Available at: <https://doi.org/10.1038/s41467-026-74859-3>.
- Roberts, D.R., Bahn, V., Ciuti, S. et al. (2017) 'Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure', *Ecography*, 40(8), pp. 913-929. Available at: <https://doi.org/10.1111/ecog.02881>.
- Salazar, D., Jackson, D., Guendon, J.L., Salinas, H., Morata, D., Figueroa, V., Manríquez, G. and Castro, V. (2011) 'Early evidence (ca. 12,000 BP) for iron oxide mining on the Pacific coast of South America', *Current Anthropology*, 52(3), pp. 463–475. Available at: <https://doi.org/10.1086/659426>.
- Simionato, J.I., Guasselli, L.A., Ruiz, L.F.C. et al. (2021) 'Identification of artisanal mining sites in the Amazon rainforest using GEOBIA and data mining techniques', *Remote Sensing Applications: Society and Environment*, 24, 100633. Available at: <https://doi.org/10.1016/j.rsase.2021.100633>.
- Sonter, L.J., Herrera, D., Barrett, D.J. et al. (2017) 'Mining drives extensive deforestation in the Brazilian Amazon', *Nature Communications*, 8, 1013. Available at: <https://doi.org/10.1038/s41467-017-00557-w>.
- Srivastava, V. and Jha, P.K. (2025) 'Environmental impacts of coal mining and mitigation measures: a review', *Environmental Monitoring and Assessment*, 197, article 545. Available at: <https://doi.org/10.1007/s10661-025-13939-1>.

- Swenson, J.J., Carter, C.E., Domec, J.-C. and Delgado, C.I. (2011) 'Gold mining in the Peruvian Amazon: global prices, deforestation, and mercury imports', PLOS ONE, 6(4), e18875. Available at: <https://doi.org/10.1371/journal.pone.0018875>.
- Tamiminia, H., Salehi, B., Mahdianpari, M., Quackenbush, L., Adeli, S. and Brisco, B. (2020) 'Google Earth Engine for geo-big data applications: A meta-analysis and systematic review', ISPRS Journal of Photogrammetry and Remote Sensing, 164, pp. 152-170. <https://doi.org/10.1016/j.isprsjprs.2020.04.001>.
- Tang, L. and Werner, T.T. (2023) 'Global mining footprint mapped from high-resolution satellite imagery', Communications Earth & Environment, 4, 134. Available at: <https://doi.org/10.1038/s43247-023-00805-6>.
- Tarras-Wahlberg, N.H., Flachier, A., Lane, S.N. and Sangfors, O. (2001) 'Environmental impacts and metal exposure of aquatic ecosystems in rivers contaminated by small scale gold mining: the Puyango River basin, southern Ecuador', Science of the Total Environment, 278(1-3), pp. 239-261. Available at: [https://doi.org/10.1016/S0048-9697\(01\)00655-6](https://doi.org/10.1016/S0048-9697(01)00655-6).
- Thiruchittampalam, S., Banerjee, B.P., Glenn, N.F. and Raval, S. (2025) 'A systematic review of machine learning-based remote sensing data analysis for geological and mined materials characterisation', European Journal of Remote Sensing, 58(1), article 2524622. Available at: <https://doi.org/10.1080/22797254.2025.2524622>.
- Tuia, D., Persello, C. and Bruzzone, L. (2016) 'Domain adaptation for the classification of remote sensing data: An overview of recent advances', IEEE Geoscience and Remote Sensing Magazine, 4(2), pp. 41-57. <https://doi.org/10.1109/MGRS.2016.2548504>.
- U.S. Geological Survey (2025) 'How do we extract minerals?'. Available at: <https://www.usgs.gov/faqs/how-do-we-extract-minerals> (Accessed: 4 September 2026).
- UNDP, World Economic Forum, Columbia Center on Sustainable Investment and Sustainable Development Solutions Network (2016) Mapping Mining to the Sustainable Development Goals: An Atlas. New York: United Nations Development Programme. Available at: <https://www.undp.org/publications/mapping-mining-sdgs-atlas> (Accessed: 4 September 2026).
- UNEP (2012) Reducing Mercury Use in Artisanal and Small-Scale Gold Mining: A Practical Guide. Nairobi: United Nations Environment Programme. Available at:

<https://www.unep.org/resources/report/reducing-mercury-use-artisanal-and-small-scale-gold-mining-practical-guide>.

UNEP (2019) Global Mercury Assessment 2018. Nairobi: United Nations Environment Programme. Available at: <https://www.unep.org/resources/publication/global-mercury-assessment-2018>.

United Nations Environment Programme and International Resource Panel (UNEP and IRP) (2024) Global Resources Outlook 2024: Bend the Trend – Pathways to a Liveable Planet as Resource Use Spikes. Nairobi: United Nations Environment Programme. Available at: https://wedocs.unep.org/bitstream/handle/20.500.11822/44901/Global-Resource-Outlook_2024.pdf.

United Nations General Assembly (2015) Transforming Our World: The 2030 Agenda for Sustainable Development. A/RES/70/1. New York: United Nations. Available at: <https://sdgs.un.org/2030agenda>.

Valavi, R., Elith, J., Lahoz-Monfort, J.J. and Guillera-Arroita, G. (2019) 'blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models', *Methods in Ecology and Evolution*, 10(2), pp. 225-232. <https://doi.org/10.1111/2041-210X.13107>.

Veiga, M.M. and Gunson, A.J. (2020) 'Gravity concentration in artisanal gold mining', *Minerals*, 10(11), 1026. Available at: <https://doi.org/10.3390/min10111026>.

Verbrugge, B. and Geenen, S. (2020) *Global Gold Production Touching Ground: Expansion, Informalization, and Technological Innovation*. Cham: Palgrave Macmillan. Available at: <https://doi.org/10.1007/978-3-030-38486-9>.

Wadoux, A.M.J.-C., Heuvelink, G.B.M., de Bruin, S. and Brus, D.J. (2021) 'Spatial cross-validation is not the right way to evaluate map accuracy', *Ecological Modelling*, 457, 109692. Available at: <https://doi.org/10.1016/j.ecolmodel.2021.109692>.

WHO (2016) *Environmental and Occupational Health Hazards Associated with Artisanal and Small-Scale Gold Mining*. Geneva: World Health Organization. Available at: <https://www.who.int/publications/i/item/9789241510271>.

World Bank Group (2021) *Creating Markets in Guinea: Generating Diversified Growth in a Resource-Rich Environment. Country Private Sector Diagnostic*. Washington, DC: World Bank Group. Available at:

<https://documents1.worldbank.org/curated/en/259311613547124114/pdf/Country-Private-Sector-Diagnostic-Creating-Markets-in-Guinea-Generating-Diversified-Growth-in-a-Resource-Rich-Environment.pdf>.